

Project #3 Report: Log Monitoring and Analysis

Objective

The objective of this project is to monitor and analyze Linux system logs using journalctl. Log monitoring helps identify security events, system errors, failed login attempts, and suspicious activities.

Tools and Technologies

- VMware Workstation
- Kali Linux
- Linux Terminal
- systemd journal (journalctl)

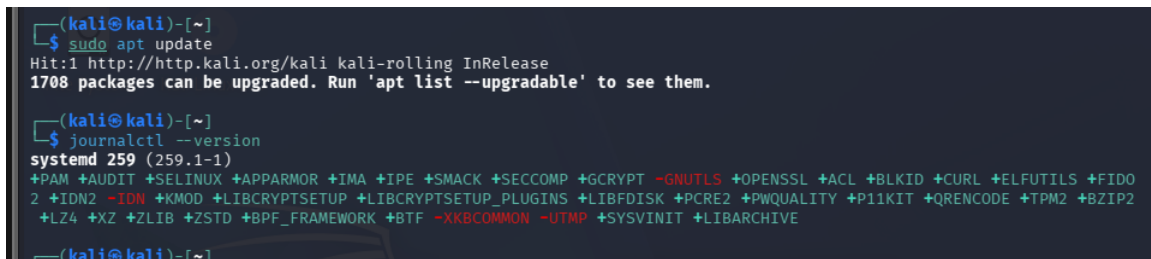
Step 1: Update Package List

Command:

```
sudo apt update
```

Description: Updates the package repository information.

Screenshot:



```
(kali@kali)-[~]
└─$ sudo apt update
Hit:1 http://http.kali.org/kali kali-rolling InRelease
1708 packages can be upgraded. Run 'apt list --upgradable' to see them.

(kali@kali)-[~]
└─$ journalctl --version
systemd 259 (259.1-1)
+PAM +AUDIT +SELINUX +APPARMOR +IMA +IPE +SMACK +SECCOMP +GCRYPT -GNUTLS +OPENSSL +ACL +BLKID +CURL +ELFUTILS +FIDO
2 +IDN2 -IDN +KMOD +LIBCRYPTSETUP +LIBCRYPTSETUP_PLUGINS +LIBFDISK +PCRE2 +PWQUALITY +P11KIT +QRENCODE +TPM2 +BZIP2
+LZ4 +XZ +ZLIB +ZSTD +BPF_FRAMEWORK +BTF -XKBCOMMON -UTMP +SYSVINIT +LIBARCHIVE
```

Step 2: View Recent Logs

Command:

```
sudo journalctl -n 50
```

Description: Displays the latest 50 log entries.

Screenshot:

```

└─$ sudo journalctl -n 50
Jul 08 06:35:24 kali sudo[5561]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
Jul 08 06:35:24 kali sudo[5561]: pam_unix(sudo:session): session closed for user root
Jul 08 06:35:58 kali sudo[5891]:    kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/sbin/ufw allow 5
Jul 08 06:35:58 kali sudo[5891]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
Jul 08 06:35:59 kali sudo[5891]: pam_unix(sudo:session): session closed for user root
Jul 08 06:36:23 kali sudo[6138]:    kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/sbin/ufw allow 22
Jul 08 06:36:23 kali sudo[6138]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
Jul 08 06:36:23 kali sudo[6138]: pam_unix(sudo:session): session closed for user root
Jul 08 06:36:59 kali sudo[6495]:    kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/sbin/ufw allow 80
Jul 08 06:36:59 kali sudo[6495]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
Jul 08 06:37:00 kali sudo[6495]: pam_unix(sudo:session): session closed for user root
Jul 08 06:37:21 kali sudo[6709]:    kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/sbin/ufw allow 4
Jul 08 06:37:21 kali sudo[6709]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
Jul 08 06:37:21 kali sudo[6709]: pam_unix(sudo:session): session closed for user root
Jul 08 06:37:39 kali sudo[6911]:    kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/sbin/ufw deny 23
Jul 08 06:37:39 kali sudo[6911]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
Jul 08 06:37:39 kali sudo[6911]: pam_unix(sudo:session): session closed for user root
Jul 08 06:38:08 kali sudo[7195]:    kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/sbin/ufw status
Jul 08 06:38:08 kali sudo[7195]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
Jul 08 06:38:08 kali sudo[7195]: pam_unix(sudo:session): session closed for user root
Jul 08 06:39:01 kali CRON[7635]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
Jul 08 06:39:01 kali CRON[7637]: (root) CMD ( [ -x /usr/lib/php/sessionclean ] && if [ ! -d /run/systemd/system ]
Jul 08 06:39:01 kali CRON[7635]: pam_unix(cron:session): session closed for user root
Jul 08 06:39:10 kali systemd[1]: Starting phpsessionclean.service - Clean php session files...
Jul 08 06:39:10 kali systemd[1]: phpsessionclean.service: Deactivated successfully.
Jul 08 06:39:10 kali systemd[1]: Finished phpsessionclean.service - Clean php session files.
Jul 08 06:43:58 kali systemd[1]: Starting systemd-tmpfiles-clean.service - Cleanup of Temporary Directories...
Jul 08 06:43:58 kali systemd-tmpfiles[10104]: /usr/lib/tmpfiles.d/legacy.conf:14: Duplicate line for path "/run/lo
Jul 08 06:43:58 kali systemd[1]: systemd-tmpfiles-clean.service: Deactivated successfully.
Jul 08 06:43:58 kali systemd[1]: Finished systemd-tmpfiles-clean.service - Cleanup of Temporary Directories.
Jul 08 06:44:17 kali NetworkManager[722]: <info> [1783507457.2158] dhcp4 (eth0): state changed new lease, address
Jul 08 06:44:17 kali dbus-daemon[652]: [system] Activating via systemd: service name='org.freedesktop.nm_dispatcher
Jul 08 06:44:17 kali systemd[1]: Starting NetworkManager-dispatcher.service - Network Manager Script Dispatcher Se
Jul 08 06:44:17 kali dbus-daemon[652]: [system] Successfully activated service 'org.freedesktop.nm_dispatcher'
Jul 08 06:44:17 kali systemd[1]: Started NetworkManager-dispatcher.service - Network Manager Script Dispatcher Ser
Jul 08 06:44:27 kali systemd[1]: NetworkManager-dispatcher.service: Deactivated successfully.
Jul 08 06:45:01 kali CRON[10663]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
Jul 08 06:45:01 kali CRON[10665]: (root) CMD (command -v debian-sa1 > /dev/null && debian-sa1 1 1)
Jul 08 06:45:01 kali CRON[10663]: pam_unix(cron:session): session closed for user root
Jul 08 06:53:15 kali sudo[14833]:    kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/cat /var/lo
Jul 08 06:53:15 kali sudo[14833]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
Jul 08 06:53:15 kali sudo[14833]: pam_unix(sudo:session): session closed for user root
Jul 08 06:54:20 kali sudo[15424]:    kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/apt update
Jul 08 06:54:20 kali sudo[15424]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)

```

```

└─$ sudo journalctl -n 50
Jul 08 06:35:24 kali sudo[5561]: pam_unix(sudo:session): session opened for user root
Jul 08 06:35:24 kali sudo[5561]: pam_unix(sudo:session): session closed for user root
Jul 08 06:35:58 kali sudo[5891]:    kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/sbin/ufw allow ssh
Jul 08 06:35:58 kali sudo[5891]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
Jul 08 06:35:59 kali sudo[5891]: pam_unix(sudo:session): session closed for user root
Jul 08 06:36:23 kali sudo[6138]:    kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/sbin/ufw allow 22
Jul 08 06:36:23 kali sudo[6138]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
Jul 08 06:36:23 kali sudo[6138]: pam_unix(sudo:session): session closed for user root
Jul 08 06:36:59 kali sudo[6495]:    kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/sbin/ufw allow 80
Jul 08 06:36:59 kali sudo[6495]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
Jul 08 06:37:00 kali sudo[6495]: pam_unix(sudo:session): session closed for user root
Jul 08 06:37:21 kali sudo[6709]:    kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/sbin/ufw allow 443
Jul 08 06:37:21 kali sudo[6709]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
Jul 08 06:37:21 kali sudo[6709]: pam_unix(sudo:session): session closed for user root
Jul 08 06:37:39 kali sudo[6911]:    kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/sbin/ufw deny 23
Jul 08 06:37:39 kali sudo[6911]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
Jul 08 06:37:39 kali sudo[6911]: pam_unix(sudo:session): session closed for user root
Jul 08 06:38:08 kali sudo[7195]:    kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/sbin/ufw status numbered
Jul 08 06:38:08 kali sudo[7195]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
Jul 08 06:38:08 kali sudo[7195]: pam_unix(sudo:session): session closed for user root
Jul 08 06:39:01 kali CRON[7635]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
Jul 08 06:39:01 kali CRON[7637]: (root) CMD ( [ -x /usr/lib/php/sessionclean ] && if [ ! -d /run/systemd/system ] ; then /usr/lib/php/sessionclean; fi)
Jul 08 06:39:01 kali CRON[7635]: pam_unix(cron:session): session closed for user root
Jul 08 06:39:10 kali systemd[1]: Starting phpsessionclean.service - Clean php session files...
Jul 08 06:39:10 kali systemd[1]: phpsessionclean.service: Deactivated successfully.
Jul 08 06:39:10 kali systemd[1]: Finished phpsessionclean.service - Clean php session files.
Jul 08 06:43:58 kali systemd[1]: Starting systemd-tmpfiles-clean.service - Cleanup of Temporary Directories...
Jul 08 06:43:58 kali systemd-tmpfiles[10104]: /usr/lib/tmpfiles.d/legacy.conf:14: Duplicate line for path "/run/lock", ignoring.
Jul 08 06:43:58 kali systemd[1]: systemd-tmpfiles-clean.service: Deactivated successfully.
Jul 08 06:43:58 kali systemd[1]: Finished systemd-tmpfiles-clean.service - Cleanup of Temporary Directories.
Jul 08 06:44:17 kali NetworkManager[722]: <info> [1783507457.2158] dhcp4 (eth0): state changed new lease, address:192.168.12.129
Jul 08 06:44:17 kali dbus-daemon[652]: [system] Activating via systemd: service name='org.freedesktop.nm_dispatcher' unit='dbus-org.freedesktop.nm-dispatcher.service' requested by '1.6' (uid=0 pid=722 comm="/usr/sbin/NetworkManager --
Jul 08 06:44:17 kali systemd[1]: Starting NetworkManager-dispatcher.service - Network Manager Script Dispatcher Service...
Jul 08 06:44:17 kali dbus-daemon[652]: [system] Successfully activated service 'org.freedesktop.nm_dispatcher'
Jul 08 06:44:17 kali systemd[1]: Started NetworkManager-dispatcher.service - Network Manager Script Dispatcher Service.
Jul 08 06:44:27 kali systemd[1]: NetworkManager-dispatcher.service: Deactivated successfully.
Jul 08 06:45:01 kali CRON[10663]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
Jul 08 06:45:01 kali CRON[10665]: (root) CMD (command -v debian-sa1 > /dev/null && debian-sa1 1 1)
Jul 08 06:45:01 kali CRON[10663]: pam_unix(cron:session): session closed for user root
Jul 08 06:53:15 kali sudo[14833]:    kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/cat /var/log/syslog
Jul 08 06:53:15 kali sudo[14833]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
Jul 08 06:53:15 kali sudo[14833]: pam_unix(sudo:session): session closed for user root
Jul 08 06:54:20 kali sudo[15424]:    kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/apt update
Jul 08 06:54:20 kali sudo[15424]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
Jul 08 06:54:22 kali sudo[15424]: pam_unix(sudo:session): session closed for user root

```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

Step 3: Monitor Logs in Real Time

Command:

sudo journalctl -f

Description: Monitors logs continuously. Press Ctrl+C to stop.

Screenshot:

```
(kali@kali)~$ sudo journalctl -f
Jul 08 06:59:17 kali systemd[1]: Starting NetworkManager-dispatcher.service - Network Manager Script Dispatcher Service...
Jul 08 06:59:17 kali dbus-daemon[652]: [system] Successfully activated service 'org.freedesktop.nm_dispatcher'
Jul 08 06:59:17 kali systemd[1]: Started NetworkManager-dispatcher.service - Network Manager Script Dispatcher Service.
Jul 08 06:59:27 kali systemd[1]: NetworkManager-dispatcher.service: Deactivated successfully.
Jul 08 07:02:32 kali sudo[16050]: pam_unix(sudo:session): session closed for user root
Jul 08 07:03:57 kali sudo[20526]:      kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/tail -f /var/log/syslog
Jul 08 07:03:57 kali sudo[20526]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
Jul 08 07:03:57 kali sudo[20526]: pam_unix(sudo:session): session closed for user root
Jul 08 07:04:24 kali sudo[20766]:      kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/journalctl -f
Jul 08 07:04:24 kali sudo[20766]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
^C
```

Step 4: View Current Boot Logs

Command:

`sudo journalctl -b`

Description: Displays logs generated since the current boot.

Screenshot:

```
(kali@kali)~$ sudo journalctl -b
Jul 08 06:29:07 kali kernel: Linux version 6.18.12+kali-amd64 (devel@kali.org) (x86_64-linux-gnu-gcc-15 (Debian 15.0.0-1)) #1 SMP PREEMPT_DYNAMIC Mon Jul 8 06:29:07 UTC 2024; root=/dev/sda1 ro qu
Jul 08 06:29:07 kali kernel: Command line: BOOT_IMAGE=/boot/vmlinuz-6.18.12+kali-amd64 root=/dev/sda1 ro qu
Jul 08 06:29:07 kali kernel: BIOS-e820: [mem 0x0000000000000000-0x00000000000009e7ff] usable
Jul 08 06:29:07 kali kernel: BIOS-e820: [mem 0x00000000000009e800-0x00000000000009ffff] reserved
Jul 08 06:29:07 kali kernel: BIOS-e820: [mem 0x0000000000000a00-0x0000000000000affff] reserved
Jul 08 06:29:07 kali kernel: BIOS-e820: [mem 0x0000000000000b00-0x0000000000000bffff] usable
Jul 08 06:29:07 kali kernel: BIOS-e820: [mem 0x0000000000000c00-0x0000000000000cffff] usable
Jul 08 06:29:07 kali kernel: BIOS-e820: [mem 0x0000000000000d00-0x0000000000000dffff] reserved
Jul 08 06:29:07 kali kernel: BIOS-e820: [mem 0x0000000000000e00-0x0000000000000effff] ACPI data
Jul 08 06:29:07 kali kernel: BIOS-e820: [mem 0x0000000000000f00-0x0000000000000fffff] ACPI NVS
Jul 08 06:29:07 kali kernel: BIOS-e820: [mem 0x0000000000001000-0x00000000000010ffff] reserved
Jul 08 06:29:07 kali kernel: BIOS-e820: [mem 0x0000000000001100-0x00000000000011ffff] reserved
Jul 08 06:29:07 kali kernel: BIOS-e820: [mem 0x0000000000001200-0x00000000000012ffff] reserved
Jul 08 06:29:07 kali kernel: BIOS-e820: [mem 0x0000000000001300-0x00000000000013ffff] usable
Jul 08 06:29:07 kali kernel: NX (Execute Disable) protection: active
Jul 08 06:29:07 kali kernel: APIC: Static calls initialized
Jul 08 06:29:07 kali kernel: SMBIOS 2.7 present.
Jul 08 06:29:07 kali kernel: DMI: VMware, Inc. VMware Virtual Platform/440BX Desktop Reference Platform, BIOS 6.00
Jul 08 06:29:07 kali kernel: DMI: Memory slots populated: 1/128
Jul 08 06:29:07 kali kernel: vmware: hypercall mode: 0x02
Jul 08 06:29:07 kali kernel: Hypervisor detected: VMware
Jul 08 06:29:07 kali kernel: vmware: TSC freq read from hypervisor: 2611.209 MHz
Jul 08 06:29:07 kali kernel: vmware: Host bus clock speed read from hypervisor: 66000000 Hz
Jul 08 06:29:07 kali kernel: vmware: using clock offset of 100391030321 ns
Jul 08 06:29:07 kali kernel: tsc: Detected 2611.209 MHz processor
Jul 08 06:29:07 kali kernel: e820: update [mem 0x00000000-0x00000fff] usable ==> reserved
Jul 08 06:29:07 kali kernel: e820: remove [mem 0x000a0000-0x0000ffff] usable
Jul 08 06:29:07 kali kernel: last_pfn = 0x140000 max_arch_pfn = 0x400000000
Jul 08 06:29:07 kali kernel: MTRR map: 8 entries (5 fixed + 3 variable; max 21), built from 8 variable MTRRs
Jul 08 06:29:07 kali kernel: x86/PAT: Configuration [0-7]: WB WC UC- UC WB WP UC- WT
Jul 08 06:29:07 kali kernel: e820: update [mem 0xc0000000-0xffffffff] usable ==> reserved
Jul 08 06:29:07 kali kernel: last_pfn = 0xc00000 max_arch_pfn = 0x400000000
Jul 08 06:29:07 kali kernel: found SMP MP-table at [mem 0x000f6a70-0x000f6a7f]
Jul 08 06:29:07 kali kernel: Using GB pages for direct mapping
Jul 08 06:29:07 kali kernel: RAMDISK: [mem 0x1f41b000-0x2ba04fff]
Jul 08 06:29:07 kali kernel: ACPI: Early table checksum verification disabled
Jul 08 06:29:07 kali kernel: ACPI: RSDP 0x00000000000f5a00 000024 (v02 PTLTD )
Jul 08 06:29:07 kali kernel: ACPI: XSDT 0x00000000bfe7633 00005c (v01 INTEL 440BX 06040000 VMW 01324272)
Jul 08 06:29:07 kali kernel: ACPI: FACP 0x00000000bfe7673 0000f4 (v01 INTEL 440BX 06040000 PTL 000f4240)
Jul 08 06:29:07 kali kernel: ACPI: DSDT 0x00000000bfe76d8 015a9b (v01 PTLTD Custom 06040000 MSFT 03000001)
Jul 08 06:29:07 kali kernel: ACPI: FACS 0x00000000bfe76fc0 000040
```

Step 5: View Error Logs

Command:

`sudo journalctl -p err`

Description: Shows only error-priority log messages.

Screenshot:

Step 7: Analyze Results

Command:

Review the displayed logs

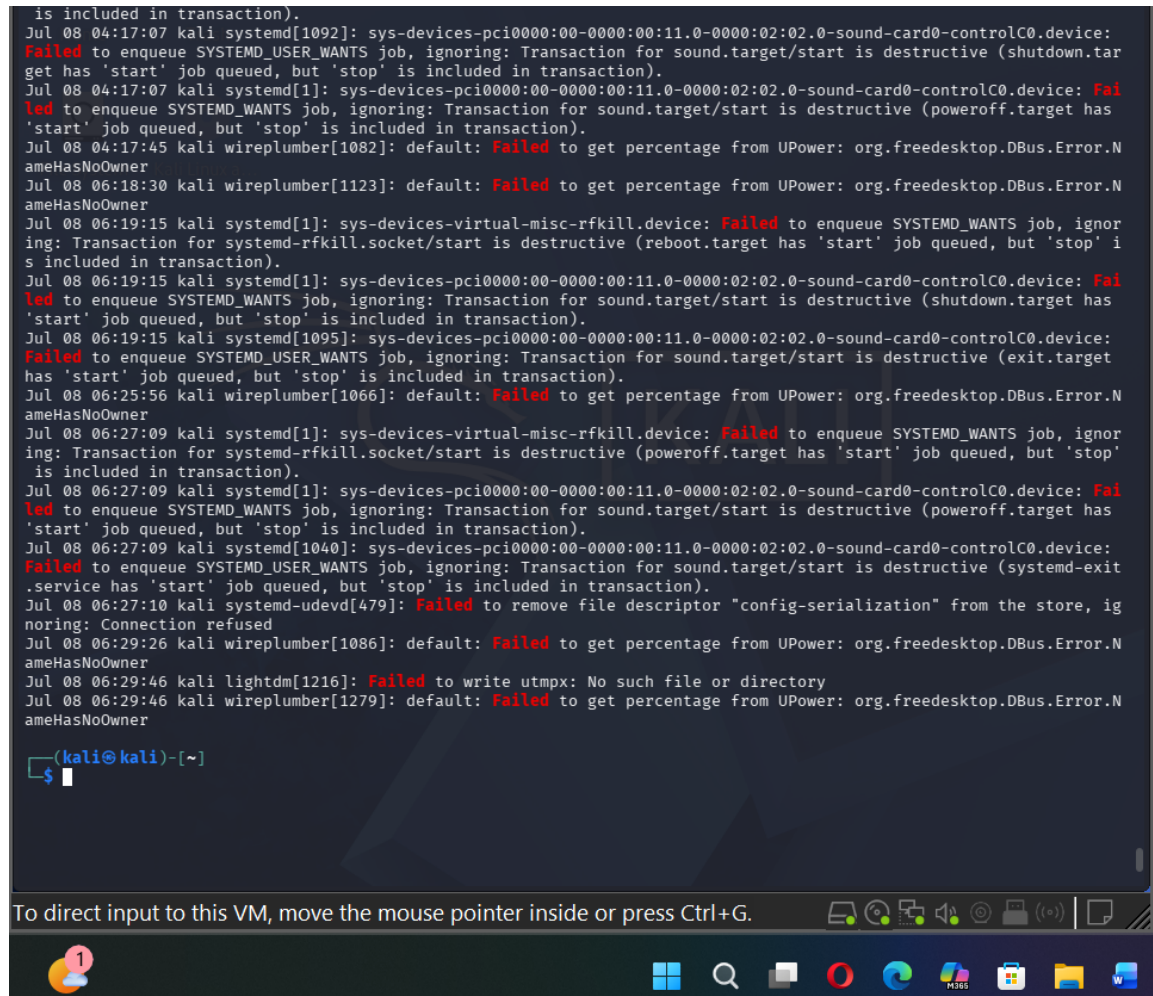
Description: Identify failed services, authentication issues, and system errors.

Screenshot:

```
is included in transaction).
Jul 08 04:17:07 kali systemd[1092]: sys-devices-pci0000:00-0000:00:11.0-0000:02:02.0-sound-card0-controlC0.device:
Failed to enqueue SYSTEMD_USER_WANTS job, ignoring: Transaction for sound.target/start is destructive (shutdown.target
has 'start' job queued, but 'stop' is included in transaction).
Jul 08 04:17:07 kali systemd[1]: sys-devices-pci0000:00-0000:00:11.0-0000:02:02.0-sound-card0-controlC0.device: Fai
led to enqueue SYSTEMD_WANTS job, ignoring: Transaction for sound.target/start is destructive (poweroff.target has
'start' job queued, but 'stop' is included in transaction).
Jul 08 04:17:45 kali wireplumber[1082]: default: Failed to get percentage from UPower: org.freedesktop.DBus.Error.N
ameHasNoOwner
Jul 08 06:18:30 kali wireplumber[1123]: default: Failed to get percentage from UPower: org.freedesktop.DBus.Error.N
ameHasNoOwner
Jul 08 06:19:15 kali systemd[1]: sys-devices-virtual-misc-rfkill.device: Failed to enqueue SYSTEMD_WANTS job, ignor
ing: Transaction for systemd-rfkill.socket/start is destructive (reboot.target has 'start' job queued, but 'stop' i
s included in transaction).
Jul 08 06:19:15 kali systemd[1]: sys-devices-pci0000:00-0000:00:11.0-0000:02:02.0-sound-card0-controlC0.device: Fai
led to enqueue SYSTEMD_WANTS job, ignoring: Transaction for sound.target/start is destructive (shutdown.target has
'start' job queued, but 'stop' is included in transaction).
Jul 08 06:19:15 kali systemd[1095]: sys-devices-pci0000:00-0000:00:11.0-0000:02:02.0-sound-card0-controlC0.device:
Failed to enqueue SYSTEMD_USER_WANTS job, ignoring: Transaction for sound.target/start is destructive (exit.target
has 'start' job queued, but 'stop' is included in transaction).
Jul 08 06:25:56 kali wireplumber[1066]: default: Failed to get percentage from UPower: org.freedesktop.DBus.Error.N
ameHasNoOwner
Jul 08 06:27:09 kali systemd[1]: sys-devices-virtual-misc-rfkill.device: Failed to enqueue SYSTEMD_WANTS job, ignor
ing: Transaction for systemd-rfkill.socket/start is destructive (poweroff.target has 'start' job queued, but 'stop'
is included in transaction).
Jul 08 06:27:09 kali systemd[1]: sys-devices-pci0000:00-0000:00:11.0-0000:02:02.0-sound-card0-controlC0.device: Fai
led to enqueue SYSTEMD_WANTS job, ignoring: Transaction for sound.target/start is destructive (poweroff.target has
'start' job queued, but 'stop' is included in transaction).
Jul 08 06:27:09 kali systemd[1040]: sys-devices-pci0000:00-0000:00:11.0-0000:02:02.0-sound-card0-controlC0.device:
Failed to enqueue SYSTEMD_USER_WANTS job, ignoring: Transaction for sound.target/start is destructive (systemd-exit
.service has 'start' job queued, but 'stop' is included in transaction).
Jul 08 06:27:10 kali systemd-udevd[479]: Failed to remove file descriptor "config-serialization" from the store, ig
noring: Connection refused
Jul 08 06:29:26 kali wireplumber[1086]: default: Failed to get percentage from UPower: org.freedesktop.DBus.Error.N
ameHasNoOwner
Jul 08 06:29:46 kali lightdm[1216]: Failed to write utmpx: No such file or directory
Jul 08 06:29:46 kali wireplumber[1279]: default: Failed to get percentage from UPower: org.freedesktop.DBus.Error.N
ameHasNoOwner

(kali@kali)~$
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.



Expected Outcome

The system displays recent logs, boot logs, error logs, and failed events. These logs help administrators monitor system health and detect security-related issues.

Conclusion

This project successfully demonstrated Linux log monitoring using journalctl. The collected logs can be analyzed to troubleshoot problems, detect suspicious activity, and improve overall system security.